Chapter A

Written tests

Two versions are offered for each chapter. The reference structure is: fluency on 6 points, an exercise worth 6 points and a problem worth 8 points.

Chapter 1 · Numerical sequences

Written test 1 · Version A

Numerical sequences **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: model, calculate, reason, communicate

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Fluency

[6 points]

1. Give the multiplier corresponding to an increase of 5%.
2. Is the sequence 12, 17, 22, 27, . . . arithmetic? Give its common difference.
3. How many terms does $u_4 + u_5 + \cdots + u_{18}$?
4. If $w_0 = 80$ and $w_{n+1} = 0{,}9w_n$, calculate w_2.
5. Write $1 + 2 + \cdots + 20$ using the symbol $\sum$.

Practice 2 · Subscriptions

[6 points] A platform has 120,000 subscribers at the start of the year and gains 11,000 each months. Let u_n the number of subscribers, in thousands, after n months.

1. Justify that (u_n) is arithmetic and state its common difference.
2. Express u_n as a function of n, then calculate u_{12}.
3. Calculate $u_1 + \cdots + u_{12}$. Interpret this sum with caution.

Practice 3 · Investment and target

[8 points] An initial capital of 650,000 Ar increases by 5 % per year. We denote C_n the capital after n years.

1. Write C_{n+1} as a function of C_n, then C_n as a function of n.
2. Calculate C_5, rounded to l ·ariary.
3. Use a calculator to find the first integer n for which C_n exceeds 1,000,000 Ar.
4. Explain why the model is geometric and not arithmetic.

Detailed solutions · teacher edition

Marking scheme. Ex. 1: 6 points spread over the fluency items. Ex. 2: 6 points. Ex. 3: 8 points.

Outline solutions. The coefficient is 1 ,05. The first flash sequence is arithmetic with common difference 5 ; the sum contains 15 terms ; $w_2 = 64$ and $\sum_{k=1}^{20} k$. For exercise 2, $u_n = 120 + 11n$, therefore $u_{12} = 252$; the requested sum uses 12 $\dfrac{u_1 + u_{12}}{2}$ and accumulates monthly states, not a number of distinct people. For exercise 3, $C_n = 650\,000 \times \left(1 + \dfrac{5}{100}\right)^n$; check the threshold with two consecutive values.

Written test 1 · Version B

Numerical sequences **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: model, calculate, reason, communicate

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Fluency

[6 points]

1. Give the multiplier corresponding to an increase of 4%.
2. Is the sequence 12, 17, 22, 27, . . . arithmetic? Give its common difference.
3. How many terms does $u_4 + u_5 + \cdots + u_{18}$?
4. If $w_0 = 80$ and $w_{n+1} = 0{,}9 w_n$, calculate w_2.
5. Write $1 + 2 + \cdots + 20$ using the symbol $\sum$.

Practice 2 · Subscriptions

[6 points] A platform has 145,000 subscribers at the start of the year and gains 9,000 each months. Let u_n the number of subscribers, in thousands, after n months.

1. Justify that (u_n) is arithmetic and state its common difference.
2. Express u_n as a function of n, then calculate u_{12}.
3. Calculate $u_1 + \cdots + u_{12}$. Interpret this sum with caution.

Practice 3 · Investment and target

[8 points] An initial capital of 720,000 Ar increases by 4 % per year. We denote C_n the capital after n years.

1. Write C_{n+1} as a function of C_n, then C_n as a function of n.
2. Calculate C_5, rounded to l ·ariary.
3. Use a calculator to find the first integer n for which C_n exceeds 1,000,000 Ar.
4. Explain why the model is geometric and not arithmetic.

Detailed solutions · teacher edition

Marking scheme. Ex. 1: 6 points spread over the fluency items. Ex. 2: 6 points. Ex. 3: 8 points.

Outline solutions. The coefficient is 1 ,04. The first flash sequence is arithmetic with common difference 5; the sum contains 15 terms; $w_2 = 64$ and $\sum_{k=1}^{20} k$. For exercise 2, $u_n = 145 + 9n$, therefore $u_{12} = 253$; the required sum uses 12 $\dfrac{u_1 + u_{12}}{2}$ and accumulates monthly states, not a number of people distinct. For exercise 3, $C_n = 720\,000 \times \left(1 + \dfrac{4}{100}\right)^n$; check the threshold with two values consecutive.

Chapter 2 · exponential functions

Written test 2 · Version A

Exponential functions **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: calculate, represent, model, interpret

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Powers and variations

[6 points]

1. Simplify $2^{4} \times 2^{-7}$ and $\dfrac{5^{x+3}}{5^{x}}$.
2. Write 3^{-2} in fraction form.
3. Give the variation of $f(x) = 1{,}04^{x}$ and of $g(x) = -2 \times 0{,}75^{x}$.
4. Explain why $a^{x} + a^{y} = a^{x+y}$ is false in general.

Practice 2 · Website traffic

[6 points] At launch, a website receives 2400 visits per day. Traffic increases by 4% per week. We model by $V(t) = 2400 \times 1{,}04^{t}$.

1. Interpret the numbers 2400 and 1.04.
2. Calculate $V(8)$, rounded to the nearest unit.
3. Determine the first whole number of weeks for which $V(t) > 3100$.

Practice 3 · Average rate

[8 points] A company's turnover rose from 82,000 to 104,000 euros in four years.

1. Calculate the overall multiplier.
2. Write the mean annual multiplier as a power.
3. Calculate the mean annual rate, rounded to 0.01%.
4. Explain why the overall rate is not simply divided by four.

Detailed solutions · teacher edition

Marking scheme: $6 + 6 + 8$. **Solution:** $2^{4} \times 2^{-7} = 2^{-3} = \dfrac{1}{8}$, $\dfrac{5^{x+3}}{5^{x}} = 125$, $3^{-2} = \dfrac{1}{9}$. f is increasing and g increasing. In the problem, calculate $V(8)$ then test two consecutive integers around the threshold.

For the average rate, $q = \left(\dfrac{V_f}{V_i}\right)^{\frac{1}{4}}$ and $t = q - 1$; the multipliers multiply.

Written test 2 · Version B

Exponential functions **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: calculate, represent, model, interpret

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Powers and variations

[6 points]

1. Simplify $2^{4} \times 2^{-7}$ and $\dfrac{5^{x+3}}{5^{x}}$.
2. Write 3^{-2} in fraction form.
3. Give the variation of $f(x) = 1{,}03^{x}$ and of $g(x) = -2 \times 0{,}75^{x}$.
4. Explain why $a^{x} + a^{y} = a^{x+y}$ is false in general.

Practice 2 · Website traffic

[6 points] At launch, a website receives 1850 visits per day. Traffic increases by 3% per week. We model by $V(t) = 1850 \times 1{,}03^{t}$.

1. Interpret the numbers 1850 and 1.03.
2. Calculate $V(8)$, rounded to the nearest unit.
3. Determine the first whole number of weeks for which $V(t) > 2250$.

Practice 3 · Average rate

[8 points] A company's turnover rose from ·82 000 to ·98000 in four years.

1. Calculate the overall multiplier.
2. Write the mean annual multiplier as a power.
3. Calculate the mean annual rate, rounded to 0.01%.
4. Explain why the overall rate is not simply divided by four.

Detailed solutions · teacher edition

Marking scheme: $6 + 6 + 8$. **Solution:** $2^{4} \times 2^{-7} = 2^{-3} = \dfrac{1}{8}$, $\dfrac{5^{x+3}}{5^{x}} = 125$, $3^{-2} = \dfrac{1}{9}$. f is increasing and g increasing. In the problem, calculate $V(8)$ then test two consecutive integers around the threshold.

For the average rate, $q = \left(\dfrac{V_f}{V_i}\right)^{\frac{1}{4}}$ and $t = q - 1$; the multipliers multiply.

Chapter 3 · Decimal logarithm function

Written test 3 · Version A

Common logarithm **Duration: 1 hour Mark: /20**

Name: _____ Class: _____

Skills: calculate, solve, reason, communicate

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Knowledge and calculations

[6 points]

1. Calculate $\log(1000)$, $\log(0{,}01)$ and $\log(10^{-4})$.
2. For $x, y > 0$, expand $\log\left(\dfrac{100x^3}{y}\right)$.
3. Solve $10^{t} = 28000$ and give t to the nearest hundredth.

Practice 2 · Growth target

[6 points] A value is multiplied by 1.05 each year. We are looking for the time needed to it doubles.

1. Express the problem as an equation.
2. Solve this equation using the common logarithm.
3. Give the first whole number of years needed to reach or exceed double.
4. Check with the two relevant consecutive powers.

Practice 3 · Sound level

[8 points] The sound level is $L = 10\log\left(\dfrac{I}{I_0}\right)$ In decibels.

1. Calculate L when $I = 10^{6} I_0$.
2. For $L = 84$, express and then calculate the ratio $\dfrac{I}{I_0}$.
3. Compare intensities corresponding to 84 dB and 74 dB.

Detailed solutions · teacher edition

Marking scheme: $6 + 6 + 8$. **Solution:** The usual values are 3, $-2, -4$. Expanding gives $2 + 3\log x - \log y$. $t = \log(28000)$. For doubling, $x = \dfrac{\log 2}{\log(1{,}05)}$ then we take the integer upper and we check. $\dfrac{I}{I_0} = 10^{\frac{L}{10}}$; a difference of 10 dB corresponds to an intensity multiplied by 10.

Written test 3 · Version B

Common logarithm **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: calculate, solve, reason, communicate

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Knowledge and calculations

[6 points]

1. Calculate $\log(1000)$, $\log(0{,}01)$ and $\log(10^{-4})$.
2. For $x, y > 0$, expand $\log\left(\dfrac{100x^3}{y}\right)$.
3. Solve $10^{t} = 32000$ and give t to the nearest hundredth.

Practice 2 · Growth target

[6 points] A value is multiplied by 1.07 each year. We are looking for the time needed to it doubles.

1. Express the problem as an equation.
2. Solve this equation using the common logarithm.
3. Give the first whole number of years needed to reach or exceed double.
4. Check with the two relevant consecutive powers.

Practice 3 · Sound level

[8 points] The sound level is $L = 10\log\left(\dfrac{I}{I_0}\right)$ In decibels.

1. Calculate L when $I = 10^{6} I_0$.
2. For $L = 78$, express and then calculate the ratio $\dfrac{I}{I_0}$.
3. Compare intensities corresponding to 78 dB and 68 dB.

Detailed solutions · teacher edition

Marking scheme: $6 + 6 + 8$. **Solution:** The usual values are 3, $-2, -4$. Expanding gives $2 + 3\log x - \log y$. $t = \log(32000)$. For doubling, $x = \dfrac{\log 2}{\log(1{,}07)}$ then we take the integer upper and we check. $\dfrac{I}{I_0} = 10^{\frac{L}{10}}$; a difference of 10 dB corresponds to an intensity multiplied by 10.

Chapter 4 · Reciprocal function

Written test 4 · Version A

Reciprocal function **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: calculate, reason, represent, communicate

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Reciprocal function

[6 points] Consider $f(x) = 2 + \dfrac{16}{x}$.

1. Give its domain.
2. Calculate its derivative and study its sign.
3. Give its asymptotes and its variation on each interval of the domain.

Practice 2 · Sign table

[6 points] Solve $\dfrac{(x-4)(x+2)}{x} \geq 0$. The table must show the zeros and the value forbidden.

Practice 3 · Average cost

[8 points] For $x > 0$, an average cost is modelled by $C(x) = x + \dfrac{16}{x} + 12$.

1. Calculate $C'(x)$ and factorise its numerator.
2. Draw the variation table of C on $]0, +\infty[$.
3. Determine the value of x which minimises the average cost, and calculate this minimum.
4. Draw a consistent sketch of the curve.

Detailed solutions · teacher edition

Marking scheme: $6 + 6 + 8$. **Solution:** $D_f = \mathbb{R} \setminus \{0\}$, $f'(x) = -\dfrac{16}{x^2} < 0$, asymptotes $x = 0$ and $y = 2$. For the quotient, arrange -2, 0, 4 and exclude 0. For the cost, $C'(x) = 1 - \dfrac{16}{x^2} = \dfrac{(x-4)(x+4)}{x^2}$; minimum at $x = 4$, with value 20.

Written test 4 · Version B

Reciprocal function **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: calculate, reason, represent, communicate

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Reciprocal function

[6 points] Consider $f(x) = 2 + \dfrac{9}{x}$.

1. Give its domain.
2. Calculate its derivative and study its sign.
3. Give its asymptotes and its variation on each interval of the domain.

Practice 2 · Sign table

[6 points] Solve $\dfrac{(x-3)(x+2)}{x} \geq 0$. The table must show the zeros and the value forbidden.

Practice 3 · Average cost

[8 points] For $x > 0$, an average cost is modelled by $C(x) = x + \dfrac{9}{x} + 12$.

1. Calculate $C'(x)$ and factorise its numerator.
2. Draw the variation table of C on $]0, +\infty[$.
3. Determine the value of x which minimises the average cost, and calculate this minimum.
4. Draw a consistent sketch of the curve.

Detailed solutions · teacher edition

Marking scheme: $6 + 6 + 8$. **Solution:** $D_f = \mathbb{R} \setminus \{0\}$, $f'(x) = -\dfrac{9}{x^2} < 0$, asymptotes $x = 0$ and $y = 2$. For the quotient, arrange -2, 0, 3 and exclude 0. For the cost, $C'(x) = 1 - \dfrac{9}{x^2} = \dfrac{(x-3)(x+3)}{x^2}$; minimum at $x = 3$, with value 18.

Chapter 5 · Statistics with two quantitative variables

Written test 5 · Version A

Two-variable statistics **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: represent, model, calculate, interpret

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Construct and describe

[6 points] We observe the pairs $(1;8)$, $(2;11)$, $(3;13)$, $(4;17)$, $(5;19)$ and $(6;23)$.

1. Draw the scatter plot with correctly labelled axes.
2. Calculate the coordinates of the mean point.
3. Describe the trend and the spread without inferring causation.

Practice 2 · Using a fit

[6 points] A line of best fit is $y = 3{,}2x + 12$, established for $x \in [2, 15]$.

1. Estimate y for $x = 10$ and comment on the estimate.
2. Estimate x when $y = 60$.
3. Estimate y for $x = 40$, then explain why this result must be treated with caution.

Practice 3 · Change of variable

[8 points] Let $z = \log(y)$ and the fit of (x, z) is $z = 0{,}06x + 1{,}1$.

1. Express y as a function of x.
2. Estimate y for $x = 8$.
3. Explain the benefit of the change of variable and one limitation of extrapolation.

Detailed solutions · teacher edition

Marking scheme: $6 + 6 + 8$. **Solution:** Calculate $\bar{x}$ and $\bar{y}$ from the six values. With the line, substitute x or solve an equation; $x = 10$ is an interpolation and $x = 40$ a distant extrapolation. The last model is $y = 10^{0{,}06x + 1{,}1}$.

Written test 5 · Version B

Two-variable statistics **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: represent, model, calculate, interpret

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Construct and describe

[6 points] We observe the pairs $(1;8)$, $(2;11)$, $(3;13)$, $(4;17)$, $(5;22)$ and $(6;23)$.

1. Draw the scatter plot with correctly labelled axes.
2. Calculate the coordinates of the mean point.
3. Describe the trend and the spread without inferring causation.

Practice 2 · Using a fit

[6 points] A line of best fit is $y = 2{,}4x + 18$, established for $x \in [2, 15]$.

1. Estimate y for $x = 10$ and comment on the estimate.
2. Estimate x when $y = 60$.
3. Estimate y for $x = 40$, then explain why this result must be treated with caution.

Practice 3 · Change of variable

[8 points] Let $z = \log(y)$ and the fit of (x, z) is $z = 0{,}06x + 1{,}1$.

1. Express y as a function of x.
2. Estimate y for $x = 8$.
3. Explain the benefit of the change of variable and one limitation of extrapolation.

Detailed solutions · teacher edition

Marking scheme: $6 + 6 + 8$. **Solution:** Calculate $\bar{x}$ and $\bar{y}$ from the six values. With the line, substitute x or solve an equation; $x = 10$ is an interpolation and $x = 40$ a distant extrapolation. The last model is $y = 10^{0{,}06x + 1{,}1}$.

Chapter 6 · Conditional Probabilities

Written test 6 · Version A

Conditional probabilities **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: model, calculate, reason, interpret

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Weighted tree

[6 points] A proportion 0.35 of orders are express. The late-delivery rate is 0.05 among the express orders and 0.12 among the others.

1. Define the events E and R, then build the complete weighted tree.
2. Calculate $P(E \cap R)$.
3. Calculate $P(R)$ by the law of total probability.
4. Calculate $P_R(E)$ and interpret.

Practice 2 · Independence

[6 points] In another study, $P(A) = 0{,}4$, $P(B) = 0.25$ and $P(A \cap B) = 0{,}10$.

1. Calculate $P(A)P(B)$.
2. Are the events independent? Justify.
3. Calculate $P_A(B)$ and confirm the conclusion.

Practice 3 · Table of counts

[8 points] Out of 800 customers, 320 have a premium card; 96 premium customers and 72 non-premium customers have cancelled. Build the complete table then calculate the probability of being premium given that the customer has cancelled.

Detailed solutions · teacher edition

Marking scheme: $8 + 6 + 6$. **Solution:** $P(R) = P(E)P_E(R) + P(\overline{E})P_{\overline{E}}(R)$ then $P_R(E) = \dfrac{P(E \cap R)}{P(R)}$. For independence, compare the intersection with the product 0 ,10. In the table, there are 168 cancellations, so $P_R(P) = \dfrac{96}{168} = \dfrac{4}{7}$.

Written test 6 · Version B

Conditional probabilities **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: model, calculate, reason, interpret

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Weighted tree

[6 points] A proportion 0.45 of orders are express. The late-delivery rate is 0.08 among the express orders and 0.15 among the others.

1. Define the events E and R, then build the complete weighted tree.
2. Calculate $P(E \cap R)$.
3. Calculate $P(R)$ by the law of total probability.
4. Calculate $P_R(E)$ and interpret.

Practice 2 · Independence

[6 points] In another study, $P(A) = 0{,}4$, $P(B) = 0.25$ and $P(A \cap B) = 0{,}14$.

1. Calculate $P(A)P(B)$.
2. Are the events independent? Justify.
3. Calculate $P_A(B)$ and confirm the conclusion.

Practice 3 · Table of counts

[8 points] Out of 800 customers, 320 have a premium card; 96 premium customers and 72 non-premium customers have cancelled. Build the complete table then calculate the probability of being premium given that the customer has cancelled.

Detailed solutions · teacher edition

Marking scheme: $8 + 6 + 6$. **Solution:** $P(R) = P(E)P_E(R) + P(\overline{E})P_{\overline{E}}(R)$ then $P_R(E) = \dfrac{P(E \cap R)}{P(R)}$. For independence, compare the intersection with the product 0 ,10. In the table, there are 168 cancellations, so $P_R(P) = \dfrac{96}{168} = \dfrac{4}{7}$.

Chapter 7 · Discrete random variables and the binomial distribution

Written test 7 · Version A

Random variables and binomial distribution **Duration: 1 hour Mark: /20**
Name: _____ Class: _____

Skills: model, calculate, reason, interpret

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Expectation of a game

[6 points] A game costs 3 euros. The gross winnings are 0, 5 or 15 euros with probabilities 0,65, 0,25 and 0,10.

1. Construct the distribution of the net gain G.
2. Calculate $E(G)$.
3. Is the game favourable to the player? Interpret without claiming what will happen in one game.

Practice 2 · Binomial distribution

[6 points] A customer responds favourably with probability 0.30. An adviser contacts independently 10 customers. Let X the number of favourable answers.

1. Justify the distribution of X and give its parameters.
2. Calculate $P(X = 2)$.
3. Calculate $P(X = 0)$ then $P(X \geq 1)$.
4. Calculate and interpret $E(X)$.

Practice 3 · Pascal's triangle

[8 points]

1. Write the line $n = 6$ of Pascal's triangle.
2. Read $\binom{6}{2}$ and $\binom{6}{4}$.
3. Explain the role of $\binom{n}{k}$ in the binomial formula.

Detailed solutions · teacher edition

Marking scheme: $6 + 8 + 6$. **Solution:** The net gain is $-3, 2, 12$; $E(G) = -1{,}95 + 0{,}50 + 1{,}20 = -0{,}25$ euro. $X \sim \mathcal{B}(10, 0{,}30)$; apply the formula, use $1 - P(X = 0)$ for "at least one", and $E(X) = np$. The line $n = 6$ is 1, 6, 15, 20, 15, 6, 1.

Written test 7 · Version B

Random variables and binomial distribution **Duration: 1 hour Mark: /20**

Name: _____ Class: _____

Skills: model, calculate, reason, interpret

A calculator is allowed unless stated otherwise. All working must be legible and results interpreted in context.

Practice 1 · Expectation of a game

[6 points] A game costs 3 euros. The gross winnings are 0, 5 or 15 euros with probabilities 0,65, 0,25 and 0,10.

1. Construct the distribution of the net gain G.
2. Calculate $E(G)$.
3. Is the game favourable to the player? Interpret without claiming what will happen in one game.

Practice 2 · Binomial distribution

[6 points] A customer responds favourably with probability 0.25. An adviser contacts independently 8 customers. Let X the number of favourable answers.

1. Justify the distribution of X and give its parameters.
2. Calculate $P(X = 2)$.
3. Calculate $P(X = 0)$ then $P(X \geq 1)$.
4. Calculate and interpret $E(X)$.

Practice 3 · Pascal's triangle

[8 points]

1. Write the line $n = 6$ of Pascal's triangle.
2. Read $\binom{6}{2}$ and $\binom{6}{4}$.
3. Explain the role of $\binom{n}{k}$ in the binomial formula.

Detailed solutions · teacher edition

Marking scheme: $6 + 8 + 6$. **Solution:** The net gain is $-3, 2, 12$; $E(G) = -1{,}95 + 0{,}50 + 1{,}20 = -0{,}25$ euro. $X \sim \mathcal{B}(8, 0{,}25)$; apply the formula, use $1 - P(X = 0)$ for "at least one", and $E(X) = np$. The line $n = 6$ is 1, 6, 15, 20, 15, 6, 1.